# FastSpeech 2-Based Text-to-Speech Synthesis

English (Svarah) and Telugu (IndicTTS Telugu)

Technical Report — Implementation Status Snapshot

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**STATUS AND SCOPE: This report documents the work actually completed in the repository at the time of the initial GitHub push. It is not a claim that the complete FastSpeech 2 training and evaluation pipeline has been finished. Model training, final duration supervision, checkpoints, generated test audio, plots, and audio-quality evaluation remain pending.**

## Project objective

The assignment objective is to adapt and implement the FastSpeech 2 text-to-speech architecture separately for English and Telugu. The implementation is based on the open-source FastSpeech 2 repository by ming024 (https://github.com/ming024/FastSpeech2). The intended work includes language-specific text processing, dataset preparation, acoustic feature extraction, model training, inference, and evaluation using the prescribed test sentences.

# 1. Implementation overview

The repository contains separate English and Telugu project directories, with language-specific configuration and preprocessing assets. The upstream FastSpeech 2 codebase was used as the starting point. The current snapshot concentrates on dataset preparation, text representation, and exploratory alignment work. It does not yet provide a complete end-to-end runnable training pipeline for either language.

### Repository structure

| Path | Purpose / current contents |
|---|---|
| English/ | English dataset manifests, configuration, source files, and documentation. |
| Telugu/ | Telugu dataset manifests, character vocabulary, alignment review assets, configuration, source files, and documentation. |
| Results/ | Reserved for training curves, generated audios, and variance-adaptor plots; final results are not yet present. |
| Report/ | Reserved for the final technical report. |
| README.md | Project overview, current progress, and limitations. |
| LICENSE-FASTSPEECH2-UPSTREAM | Upstream FastSpeech 2 MIT license copy. |
| THIRD_PARTY_NOTICES.md | Attribution and upstream repository reference. |

# 2. Dataset preparation

## 2.1 English — Svarah

The Hugging Face dataset ai4bharat/Svarah was scanned in streaming mode. The inspected split contained 6,656 recordings with approximately 9.614 hours of audio and 117 speakers. A deterministic duration-based selection using seed 42 produced 3,453 training recordings (approximately 5 hours) and 695 validation recordings (approximately 1 hour). The remaining 2,508 recordings were not included in these two subsets. The train and validation record IDs were checked for overlap; none was found.

Audio was extracted successfully for all selected records and resampled from 16 kHz to 22,050 Hz. The metadata includes record ID, transcript, duration, gender, speaker native-language metadata, age group, and audio path. The primary_language field describes speaker native language; it should not be interpreted as the language spoken in the recording.

English transcripts were phonemized using g2p-en 2.1.0. The full selected set was processed: 3,453 training and 695 validation transcripts, with no reported phonemization errors. The phonemized manifests are included in the working data preparation, while repository manifests omit local audio paths.

## 2.2 Telugu — IndicTTS Telugu

The Hugging Face dataset SPRINGLab/IndicTTS_Telugu was scanned in streaming mode. The inspected training split contained 8,576 recordings with approximately 36.712 hours of audio. A deterministic duration-based selection using seed 42 produced 1,168 training recordings (approximately 5 hours) and 199 validation recordings (approximately 1 hour), with no record-ID overlap. The remaining 7,209 records were not selected for these subsets.

All 1,367 selected audio files were extracted successfully and resampled to 22,050 Hz. Original audio in the inspected dataset was available at 16 kHz and 48 kHz. Transcript validation found no missing or empty transcripts in the selected training set. Unicode NFC normalization was checked. The character inventory contains 71 observed characters, with padding and unknown tokens bringing the vocabulary size to 73. A few Latin 'o' characters were observed in Telugu transcripts and were preserved rather than silently substituted.

# 3. Language-specific text processing

## 3.1 English phoneme representation

The English text pipeline uses g2p-en to convert transcripts to phoneme sequences compatible with the upstream FastSpeech 2 text interface. A symbol compatibility check was performed against the repository text symbols. Twenty English records were prepared for an exploratory Montreal Forced Aligner alignment test. Custom pronunciations were added for several out-of-vocabulary words and abbreviations. These pronunciations are provisional and context-dependent.

The 20-record English alignment test produced TextGrids containing 957 phone intervals, with no remaining 'spn' intervals in that test batch. This is a limited alignment test only; it does not mean that the complete English training corpus has been aligned.

## 3.2 Telugu Unicode character tokenizer

A character-level tokenizer was created for Telugu using Unicode NFC text. Its vocabulary contains the 71 observed characters, a padding token, and an unknown token (73 entries total). Round-trip encoding and decoding was tested on a Telugu sentence, and the test preserved the original string exactly with zero unknown characters. An English regression test was also performed.

The upstream dataset loader was adapted to read the nested char_to_id mapping from the Telugu vocabulary JSON. The model configuration was also adapted to accept a language-specific text vocabulary size. These are code-level adaptations; they do not by themselves establish that Telugu acoustic model training is complete.

# 4. Telugu alignment investigation

An exploratory CTC character-alignment pipeline was tested using the model anuragshas/wav2vec2-large-xlsr-53-telugu. Character emission spans were generated for the 1,168 selected Telugu training records. Structural checks found 1,168 successful alignment outputs, no missing character spans, no character-count mismatches, no invalid timestamps, and no non-monotonic character spans. The aligned character sequence matched the reference transcript for all 1,168 records.

These structural checks do not prove that every timestamp is acoustically correct. The CTC spans mark token-emission regions and can omit blank frames and long acoustic intervals. Several recordings showed suspiciously large gaps or low alignment scores. Twelve records were flagged for manual review using a provisional score threshold, and additional candidates with large gaps were investigated. Listening checks on selected examples showed that some apparent gaps occurred during continuous speech, indicating alignment errors rather than silence.

A midpoint-based method was tested to convert adjacent CTC emission spans into character-duration partitions. It assigned leading, internal, and trailing gaps to neighboring characters and passed structural checks across the 1,168 training records: duration labels summed to the mel-frame count and no zero-duration labels were produced. However, it generated extreme outliers in some recordings, including a case where a long leading interval was assigned to the first character. Therefore, these duration labels are experimental candidates only and have not been finalized for FastSpeech 2 training.

The available Montreal Forced Aligner environment listed an English US ARPA acoustic model only. A Telugu MFA acoustic model was not available in that environment at the time of this snapshot; the English model was not used as a substitute for Telugu.

## 5. Model configuration and code adaptations

The upstream FastSpeech 2 architecture is retained as the implementation starting point. The Telugu model configuration specifies a 73-token text vocabulary, a single-speaker setup, and acoustic settings of 22,050 Hz sampling rate, 1,024-point FFT, 256-sample hop, 1,024-sample window, 80 mel channels, and an 0–8,000 Hz mel frequency range. The STFT implementation required compatibility changes for the installed Librosa API: pad_center is called with the size keyword, and the mel filter helper is called with named arguments.

The Telugu preprocessing configuration points to the IndicTTS Telugu corpus and the character vocabulary. The English preprocessing configuration still contains the original LJSpeech dataset/path settings and has not yet been fully adapted to Svarah. The original preprocessing dispatcher also does not yet support the two target datasets without additional integration work. The Telugu training configuration and complete language-specific preprocessing integration remain pending.

A pretrained LJSpeech HiFi-GAN generator was loaded successfully during an environment check. This only confirms that the checkpoint can be loaded; it does not establish suitability or output quality for Telugu or Indian-accented English. Vocoder compatibility and audio quality must be evaluated after a trained acoustic model and inference pipeline are available.

## 6. Current status of assignment deliverables

| Deliverable | Status at this snapshot |
|---|---|
| English dataset selection and 5h/1h split | Completed; no train/validation overlap. |
| English audio extraction and 22.05 kHz resampling | Completed for selected records. |
| English phonemization | Completed for selected train/validation records. |
| English forced alignment | Exploratory 20-record test only; full-corpus alignment pending. |
| Telugu dataset selection and 5h/1h split | Completed; no train/validation overlap. |
| Telugu audio extraction and 22.05 kHz resampling | Completed for selected records. |
| Telugu Unicode tokenizer | Implemented and round-trip tested. |
| Telugu CTC character alignment | Exploratory outputs generated; timestamp quality review pending. |
| Final Telugu duration labels | Not finalized; midpoint-derived labels remain experimental. |
| Complete dataset-specific preprocessing integration | Pending for both languages. |
| FastSpeech 2 training and validation | Not completed. |
| Trained checkpoints | Not available in this snapshot. |
| Inference and five test audios per language | Not completed. |
| Training/validation loss plots | Not available. |
| Duration, pitch, energy plots | Not available. |
| Listening-based audio quality evaluation | Not available. |

## 7. Limitations and next engineering steps

The following work is required before the implementation can be described as a completed FastSpeech 2 TTS system:

1. Finish dataset adapters for Svarah and IndicTTS Telugu and make preprocessing reproducible from the repository configuration.

2. Establish reliable phone/character duration supervision. Review low-confidence and long-gap alignments; do not use the current midpoint-derived Telugu labels as final labels without validation.

3. Generate and verify mel-spectrogram, pitch, energy, and duration features for train and validation data.

4. Complete separate language-specific training configurations and train FastSpeech 2 for English and Telugu. Record the actual training environment, hyperparameters, epochs, and loss values.

5. Validate a compatible vocoder and run inference on all prescribed test sentences, preserving the assignment text exactly.

6. Produce training/validation loss plots, generated audio files, and duration/pitch/energy visualizations for every test sample.

7. Perform listening-based evaluation, document pronunciation and intelligibility observations, and compare English and Telugu results based on actual outputs.

8. Add the final trained checkpoints or documented external checkpoint links, dependency versions, exact commands, and reproducibility instructions.

## 8. Reproducibility and repository notes

The public repository is the versioned record of this implementation snapshot: https://github.com/GoduguVeena/FastSpeech2-IIITH-VoiceAI. Dataset audio and large generated feature files are not included in the initial source snapshot. The manifests committed to the repository omit machine-specific local audio paths. Users reproducing the work must obtain the datasets from their respective sources and configure local paths appropriately.

The upstream FastSpeech 2 source is attributed in THIRD_PARTY_NOTICES.md, and the upstream MIT license is retained as LICENSE-FASTSPEECH2-UPSTREAM. The repository README files describe the current implementation status and identify pending work.

## Conclusion

At this snapshot, the project has completed meaningful dataset preparation for both languages, English phonemization, a Telugu character tokenizer, and exploratory alignment investigations. These results provide a foundation for the remaining FastSpeech 2 pipeline. However, model training, final acoustic feature preparation, inference, generated audio, and evaluation have not yet been completed. The repository and this report intentionally distinguish completed preparation work from pending model implementation and evaluation.

### References


1. FastSpeech 2 reference implementation: https://github.com/ming024/FastSpeech2

2. Svarah dataset: https://huggingface.co/datasets/ai4bharat/Svarah

3. IndicTTS Telugu dataset: https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu

4. Telugu XLS-R CTC model used for alignment experiments: https://huggingface.co/anuragshas/wav2vec2-large-xlsr-53-telugu